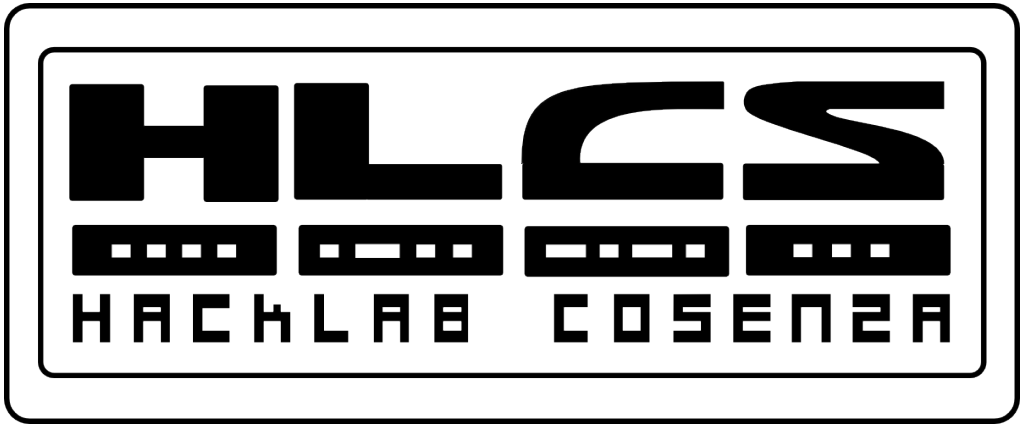

NAT & PAT

Advanced Networking: Routing & Switching 1 – Chapter 11

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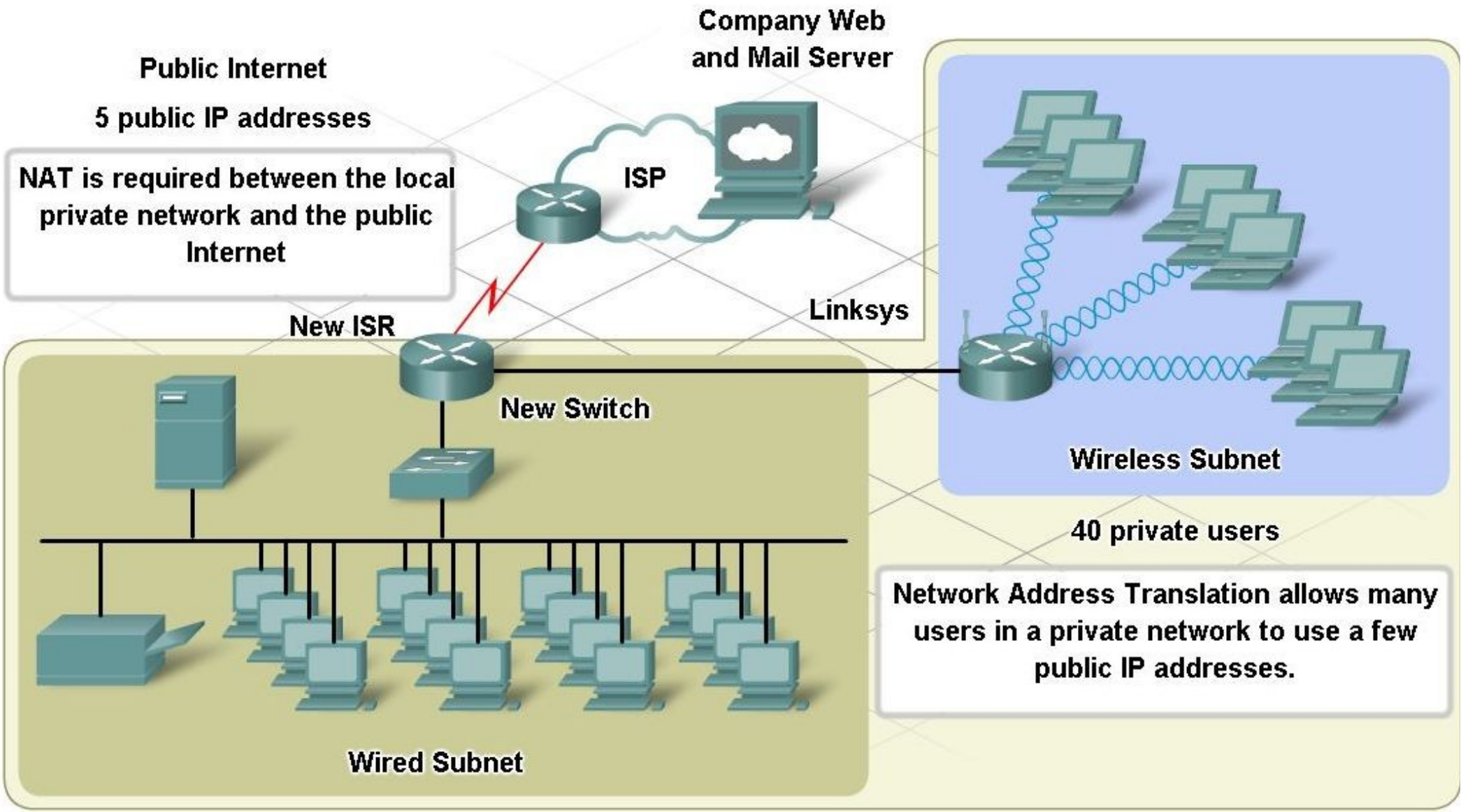
Using Network Address Translation in a Network (NAT)

- Translates addresses from one network to another
- Allows a large group of private users to access the Internet by sharing a small group of public IPs
  - comparable to a company with a few phone lines, but many extensions
- Developed to help save registered IP addresses
- Provides security to PCs, servers, devices, etc

Using Network Address Translation in a Network (NAT)



Advantages & Disadvantages of NAT

Advantages

- Reuse of IP addresses
- Sharing of a global IP address
- Security
- Scalability

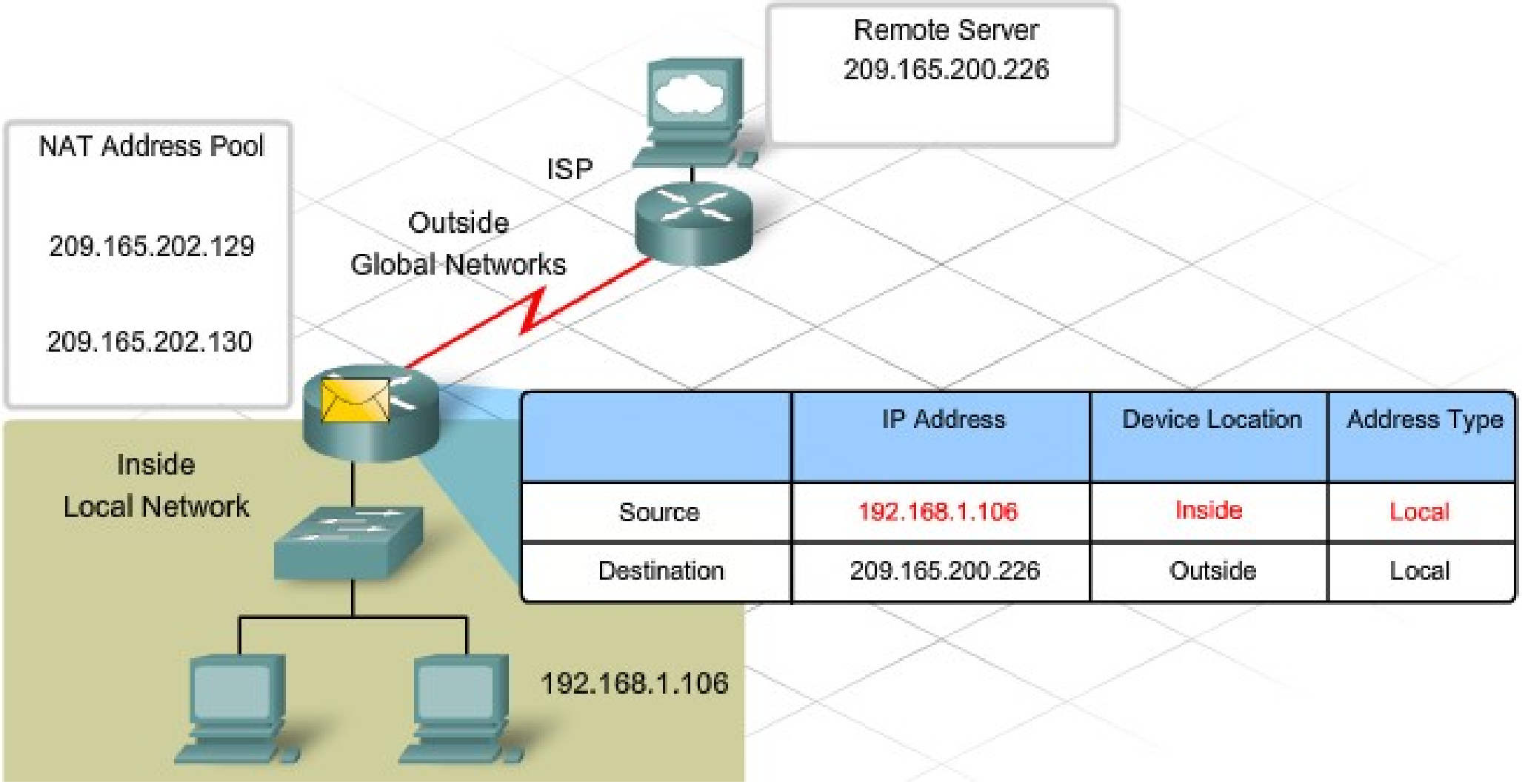
Disadvantages

- Performance is degraded.
- End-to-end functionality is degraded.
- End-to-end IP traceability is lost.
- Tunneling becomes more complicated.
- Initiating TCP connections can be disrupted.

NAT Terminology

- Inside local network
  - any network connected to a router interface that is part of the privately addressed LAN
- Outside global network
  - any network attached to a router that is external to the LAN

NAT: Inside and Outside Networks

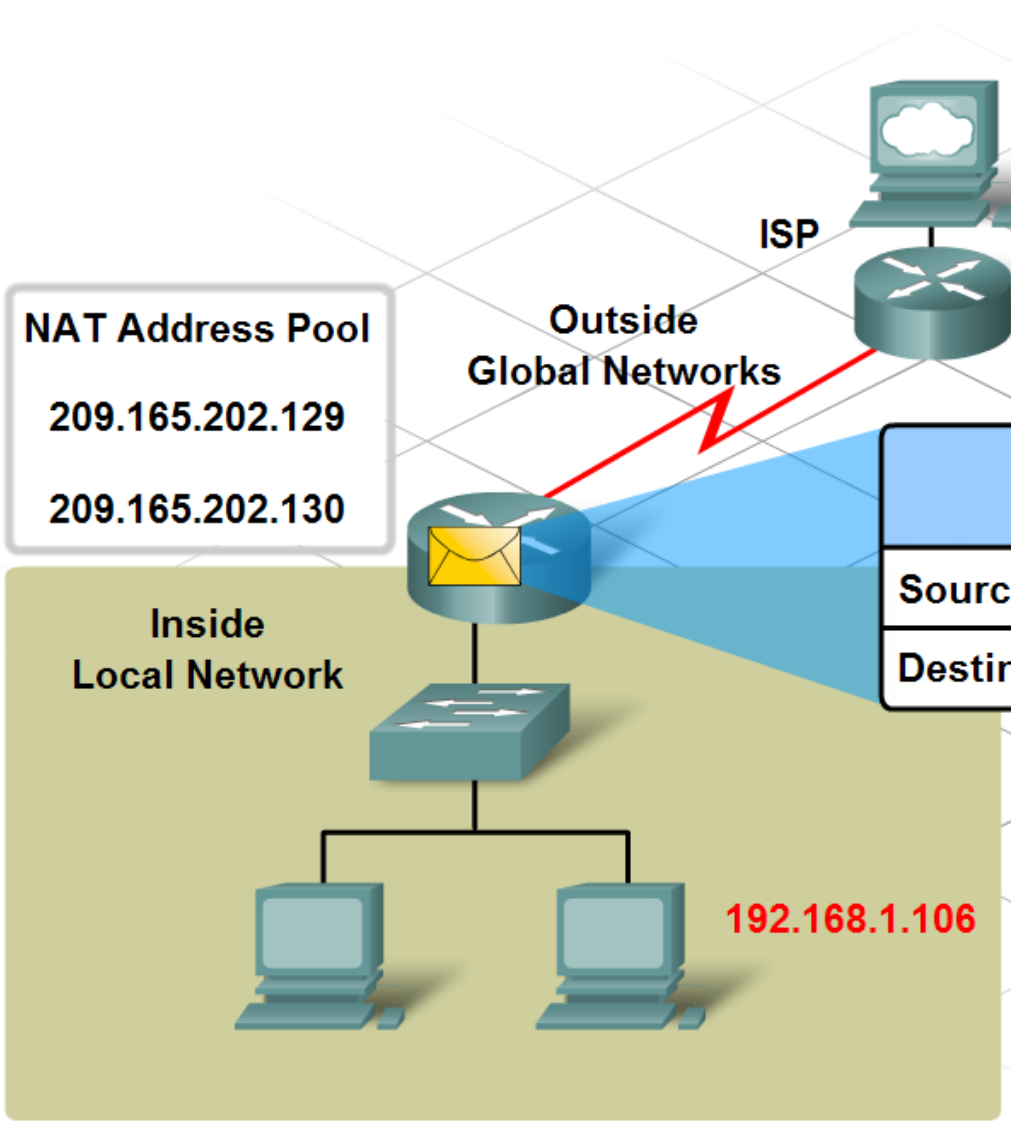


	IP Address	Device Location	Address Type
Source	192.168.1.106	Inside	Local
Destination	209.165.200.226	Outside	Local

More NAT Terminology

- Inside local address
  - Private IP address of a host on the local network
  - Must be translated to a public IP before it can travel outside
- Outside global address
  - Actual public IP address of an external host

NAT: Inside and Outside Addresses



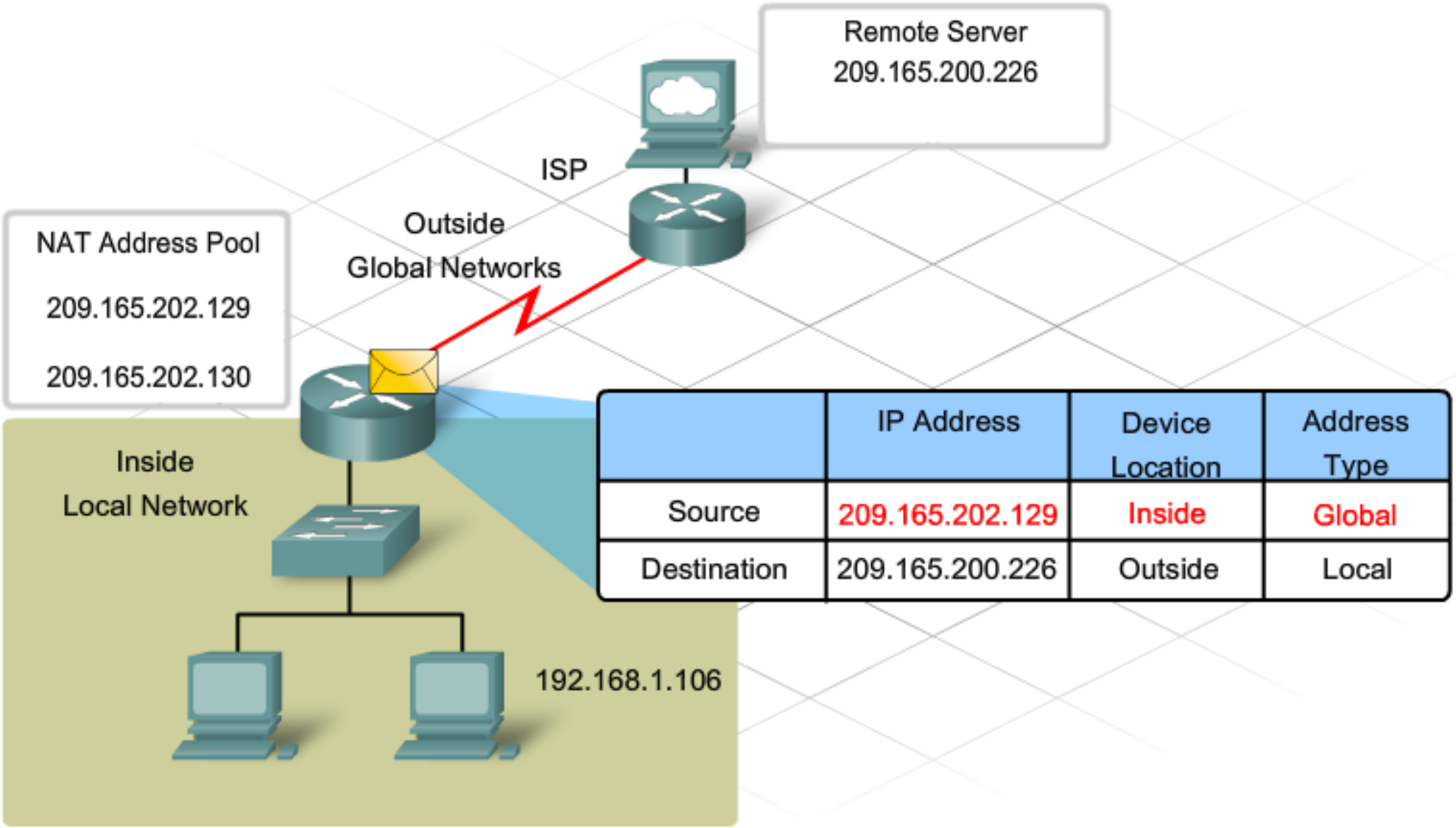
	IP Address	Device Location	Address Type
Source	209.165.200.226	Outside	Global
Destination	192.168.1.106	Inside	Local

The gateway router receives the packet and translates the destination address back to the inside private address.

More NAT Terminology - 2

- Inside global address
  - IP address of inside host as it APPEARS to the outside network
- Outside local address
  - Destination address of packet while on the local network
  - <u>Usually the same as outside global address</u>

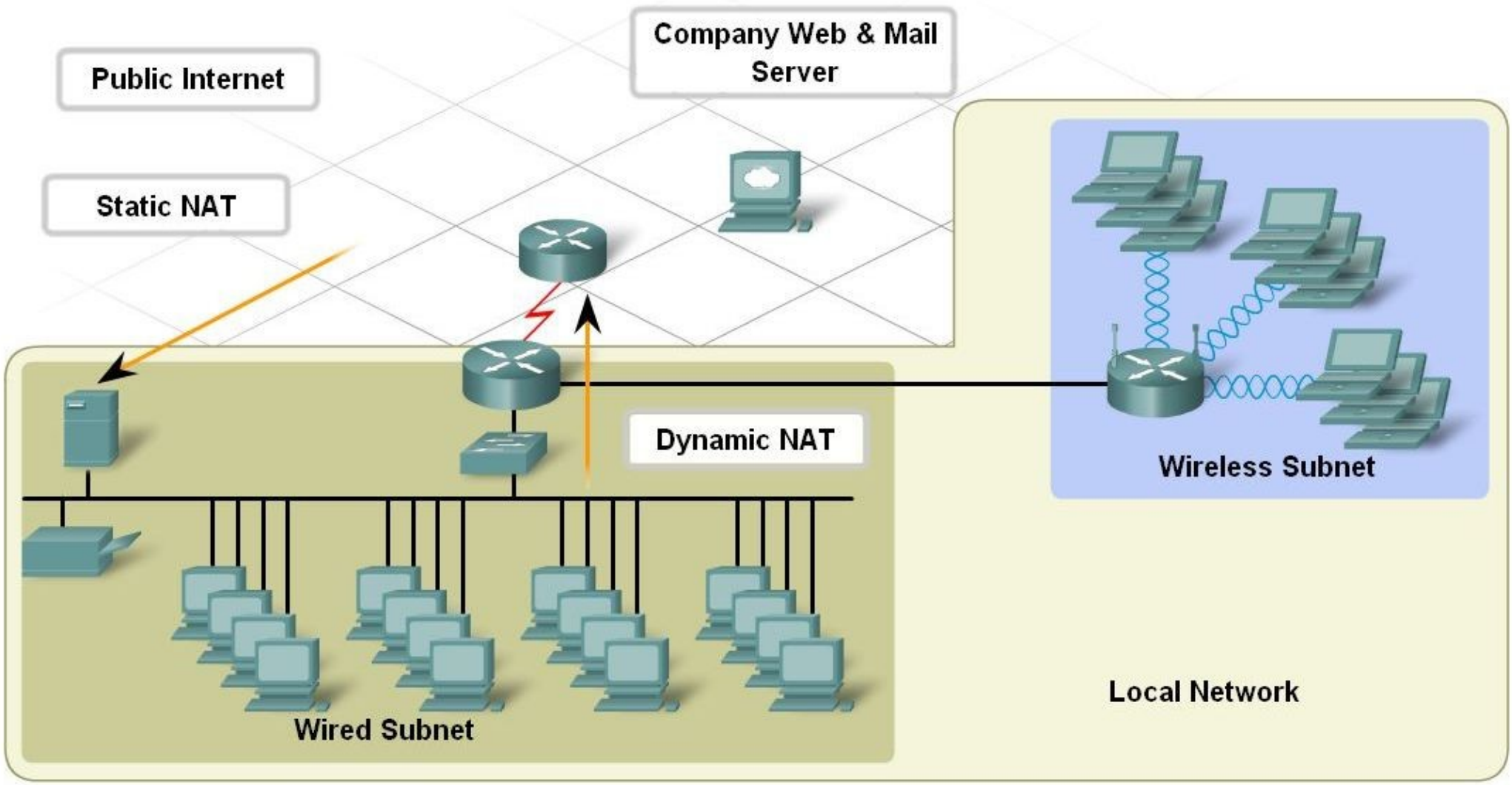
NAT: Inside and Outside Addresses - 2



	IP Address	Device Location	Address Type
Source	209.165.202.129	Inside	Global
Destination	209.165.200.226	Outside	Local

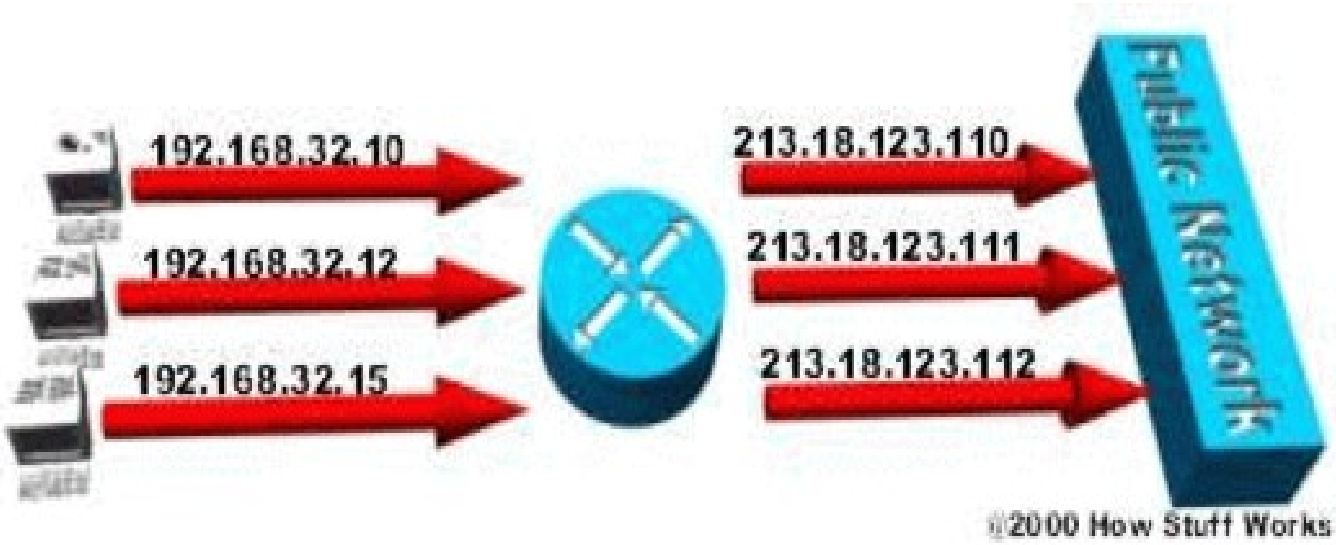
Static NAT

- Provides access to a local host from the Internet
- Allows hosts on public network to access selected hosts on a private network



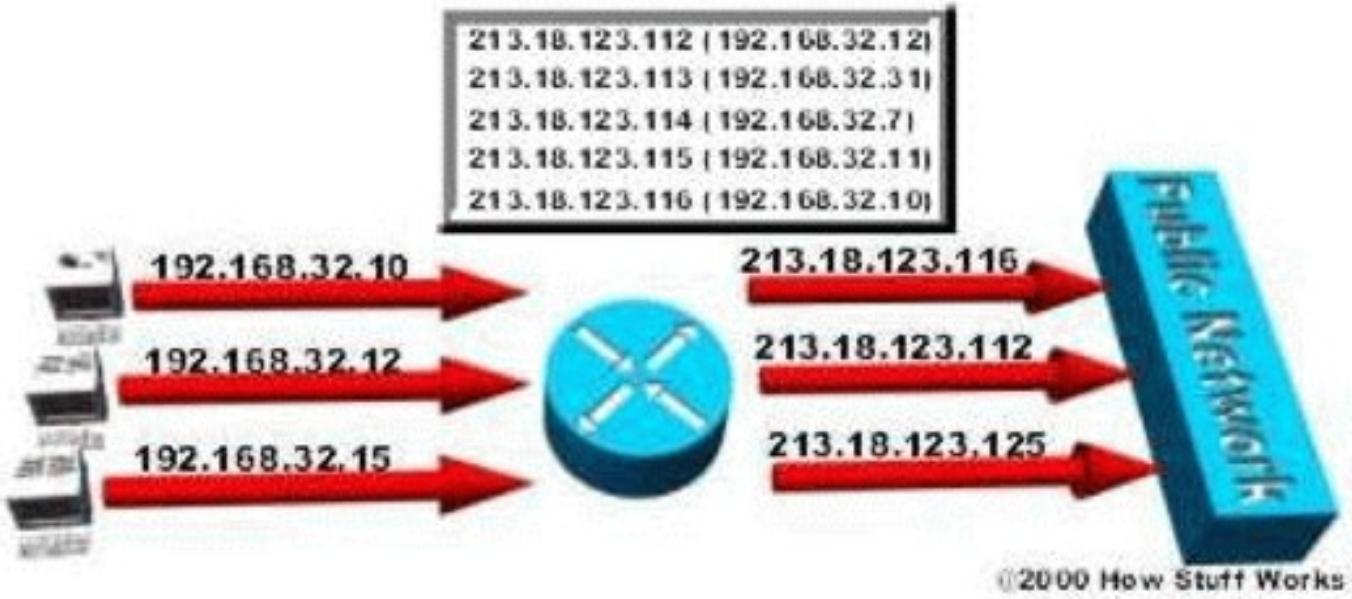
Static NAT Example

- The host with the IP address of 192.168.32.10 will always translate to 213.18.123.110.



Dynamic NAT

- Maps an unregistered IP address to a registered IP address from a group of registered IP addresses.
- The host with the IP address of 192.168.32.10 will translate to the first available address in the range of 213.18.123.10 to 213.18.123.150
- <u>Limited by number of Outside Global addresses</u>



PAT (NAT Overload)

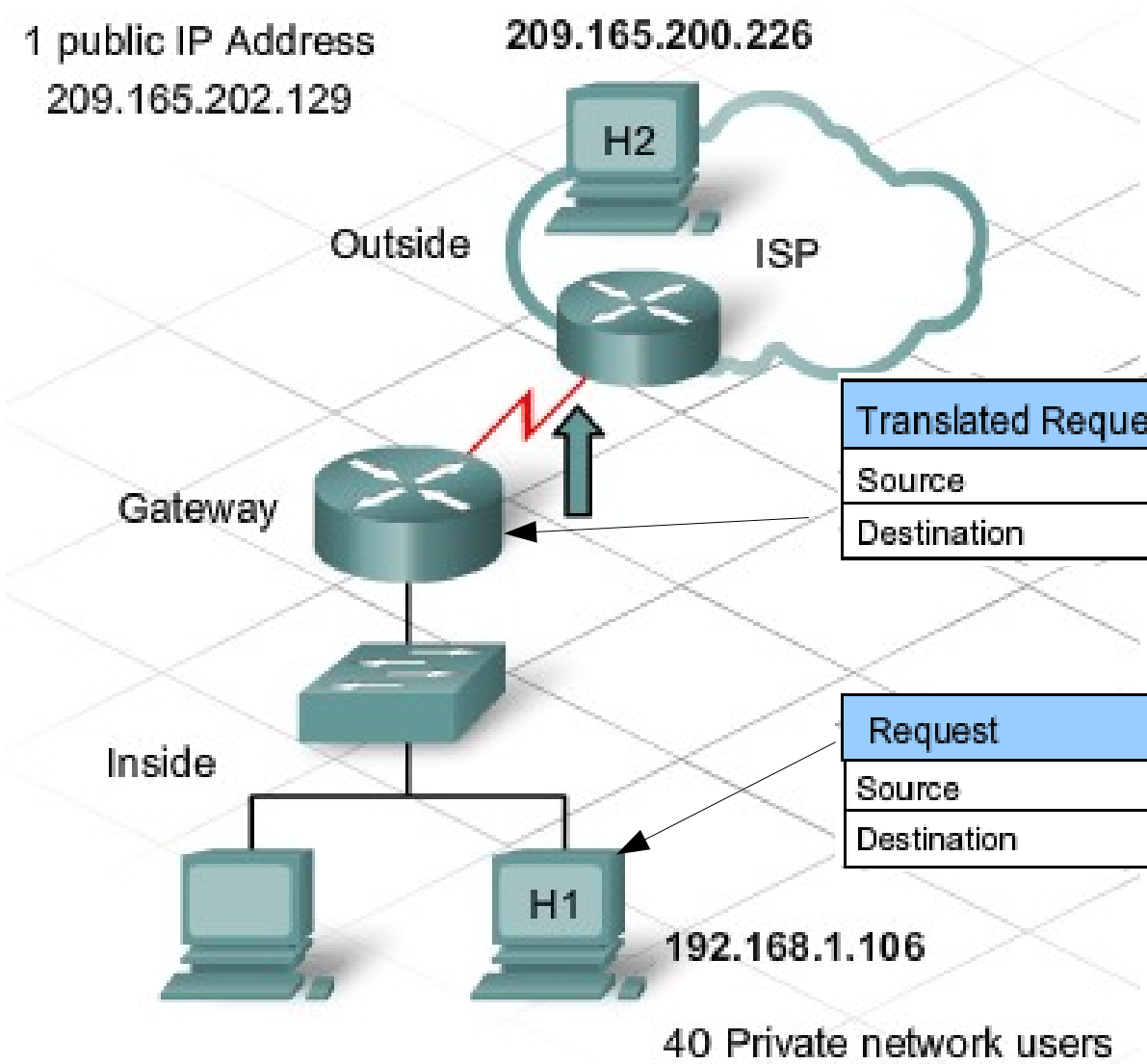
- A form of dynamic NAT that maps multiple private IP addresses to a single registered IP address by using different ports. This is known also as **PAT (Port Address Translation)**, single address NAT or port-level multiplexed NAT.
- Each computer on the private network is translated to the same IP address (213.18.123.100), but with a different port number assignment.
- <u>Outside users are unable to initiate communication to a host if using PAT</u>

PAT (Overloading)

- Uses unique source port numbers in the inside global IP address
- Uses first available port number
- These port numbers must be **above 1024**

Let's see how it works ->

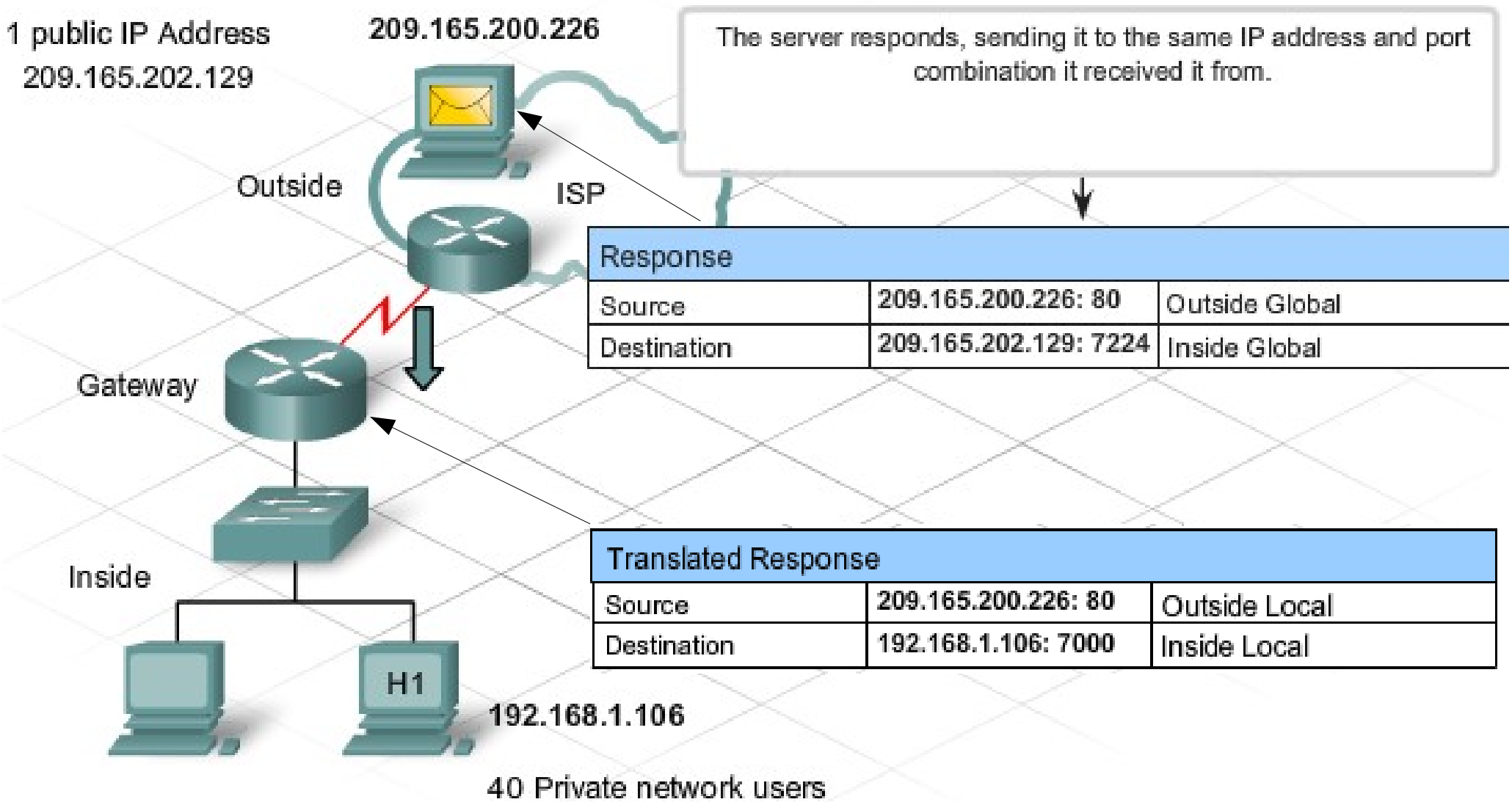
PAT: send



Translated Request		
Source	209.165.202.129: 7224	Inside Global
Destination	209.165.200.226: 80	Outside Local

Request		
Source	192.160.1.106: 7000	Inside Local
Destination	209.165.200.226: 80	Outside Local

PAT: receive



Response		
Source	209.165.200.226: 80	Outside Global
Destination	209.165.202.129: 7224	Inside Global

Translated Response		
Source	209.165.200.226: 80	Outside Local
Destination	192.168.1.106: 7000	Inside Local

Comparing Static NAT and Dynamic NAT

- Static NAT maps a single **inside local address** to a single global, or public address called the **inside global address**.
- This mapping ensures that a particular inside local address **always associates with the same** public address.
- Static NAT ensures that **outside devices** consistently reach an internal device
- Dynamic NAT uses an available **pool of Internet public addresses** and assigns them to inside local addresses.
- Dynamic NAT assigns **the first available IP** address in the pool of public addresses to an inside device.
- The NAT router manages the **translations** between the inside local addresses and the inside global addresses by **maintaining a table** that lists each address pair.

Configure static NAT

```
R1# show running-config
(*** output omitted ***)
!
ip nat inside source static 172.31.232.14  209.165.202.130
!
interface fastethernet 0/0
  ip address 172.31.232.182  255.255.255.0
  ip nat inside
!
interface serial 0/0/0
  ip address 209.165.202.1  255.255.255.0
  ip nat outside
!
```

Configure dynamic NAT

- Step 1. **Define the pool** of addresses that will be used for translation using the ip nat pool command.
  - This pool of addresses is typically a group of public addresses.
- Step 2. **Configure a standard ACL** to identify (permit) only those addresses that are to be translated.
- Step 3. **Bind the ACL to the pool**
  - The **`ip nat inside source list`** *`access-list-number`* **`pool`** *`pool-name`* command is used to bind the ACL to the pool.
  - This configuration is used by the router to identify which devices (list) receive which addresses (pool).
- Step 4. **Identify which interfaces** are **<u>inside</u>**, in relation to NAT and which interfaces are **<u>outside</u>**.

Configure dynamic NAT

```
R1# show running-config
(*** output omitted ***)
!
access-list 1 permit 172.31.232.0  0.0.0.255
ip nat pool pub-addr 209.165.202.131  209.165.202.140 netmask 255.255.255.0
ip nat inside source list 1 pool pub-addr
!
interface fastethernet 0/0
ip address 172.31.232.182  255.255.255.0
ip nat inside
!
interface serial 0/0/0
ip address 209.165.202.1  255.255.255.0
ip nat outside
!
```

Dynamic NAT maps multiple private addresses to multiple public address.

Port Address Translation

- One of the more popular variations of dynamic NAT is known as Port Address Translation (PAT), **also referred to as NAT Overload**.
- PAT dynamically translates multiple inside local addresses to a single public address.
- In PAT, the gateway router translates the **local source address and port number combination** to a single global IP address and a unique port number above 1024
- A **table in the router** contains a list of the internal IP address and port number combinations that are translated to the external address.
- Although each host translates into **the same global IP address,** the port number associated with the conversation is unique.

Configure Single Address PAT

```
R1# show running-config
(*** output omitted ***)
! Step 1: permit address to translate
access-list 1 permit 172.31.232.0 0.0.0.255
!
! Step 2: Establish dynamic source translation
ip nat inside source list 1 interface serial 0/0/0 overload
!
! Step 3: identify the inside interface
interface fastethernet 0/0
ip address 172.31.232.182 255.255.255.0
ip nat inside
!
! Step 4: identify the outside interface
interface serial 0/0/0
ip address 80.90.100.110 255.255.255.252
ip nat outside
```

Configure PAT with Address Pool

```
! Step 1: permit address to translate
access-list 1 permit 172.31.232.0 0.0.0.255
!
! Step 2: define a pool of public addresses
ip nat pool PoolName 80.90.100.226 80.90.100.240 255.255.255.224
!
! Step 3: Bind the pool with ACL
ip nat inside source list 1 pool PoolName overload
!
! Step 4: identify the inside interface
interface fastethernet 0/0
ip address 172.31.232.182 255.255.255.0
ip nat inside
!
! Step 5: identify the outside interface
interface serial 0/0/0
ip address 80.90.100.225 255.255.255.224
ip nat outside
```

Verifying PAT

```
R1# show ip nat translation
Pro  Inside global         Inside local       Outside local     Outside global
---  209.165.202.130       172.31.232.14      ------
icmp 209.165.202.131:512   172.31.232.1:512   209.165.200.1:512 209.165.200.1:512
udp  209.165.202.131:1067  172.31.232.2:1067  209.165.200.2:53  209.165.200.2:53
Tcp  209.165.202.131:1028  172.31.232.2:1028  209.165.200.3:80  209.165.200.3:80
```

```
R1# show ip nat statistics
Total active translations: 0 (0 static, 0 dynamic; 0 extended)
Outside interfaces:
  Serial 0/0/0
Inside interfaces:
  FastEthernet 0/0
Hits: 47  Misses: 0
Expired translations: 5
```

Port forwarding

- Port forwarding (sometimes referred to as tunneling) is the act of forwarding a network port from one network node to another.
- This technique allows an external user to reach a port on a private IPv4 address (inside a LAN) from the outside, through a NAT-enabled router.
- Because NAT hides internal addresses, peer-to-peer only works from the inside out where NAT can map outgoing requests against incoming replies.
- The problem is that NAT does not allow requests initiated from the outside.
- Port forwarding can be configured to identify specific ports that can be forwarded to inside hosts.

Configure Port Forwarding

Establishes static translation between an inside local address and local port and an inside global address and global port.

```
R2(config)# ip nat inside source static tcp 192.168.10.254 80
209.165.200.225 8080
```

Identifies interface serial 0/0/0 as an inside NAT interface.

```
R2(config)# interface Serial0/0/0
R2(config-if)# ip nat inside
```

Identifies interface serial 0/1/0 as the outside NAT interface.

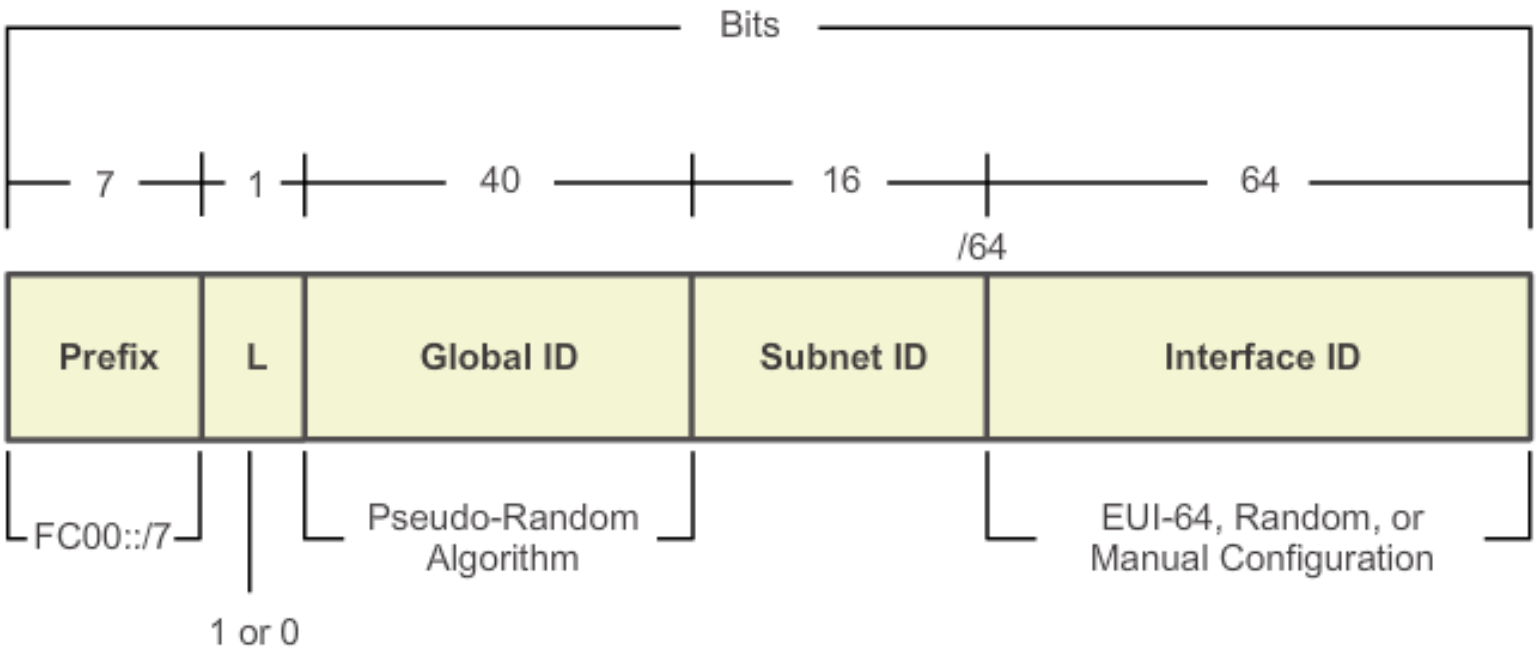
```
R2(config)# interface Serial0/1/0
R2(config-if)# ip nat outside
```

IPv6 NAT

- IPv6 with a 128-bit address provides 340 undecillion addresses.
  - Therefore, address space is not an issue.
- IPv6 was developed with the intention of making NAT unnecessary.
- However, IPv6 does implement a form of NAT.
- IPv6 includes both its own **IPv6 private address space** and **NAT**, which are implemented differently than they are for IPv4.
- IPv6 unique local addresses (ULA) are similar to RFC 1918 private addresses in IPv4.
- The intent of ULA is to provide IPv6 address space for communications within a local site.
- **NAT for IPv6 should not be used as a long term strategy, but as a temporary mechanism to assist in the migration from IPv4 to IPv6.**

IPv6 unique local addresses (ULA)

- ULA have the prefix FC00::/7, which results in a first hextet range of FC00 to FDFF.
  - The next 1 bit is set to 1 if the prefix is locally assigned. Set to 0 may be defined in the future.
- The next 40 bits is a global ID followed by a 16-bit Subnet ID.
- The first 64 bits combine to make the ULA prefix, the remaining 64 bits are the interface ID



End of lesson